



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

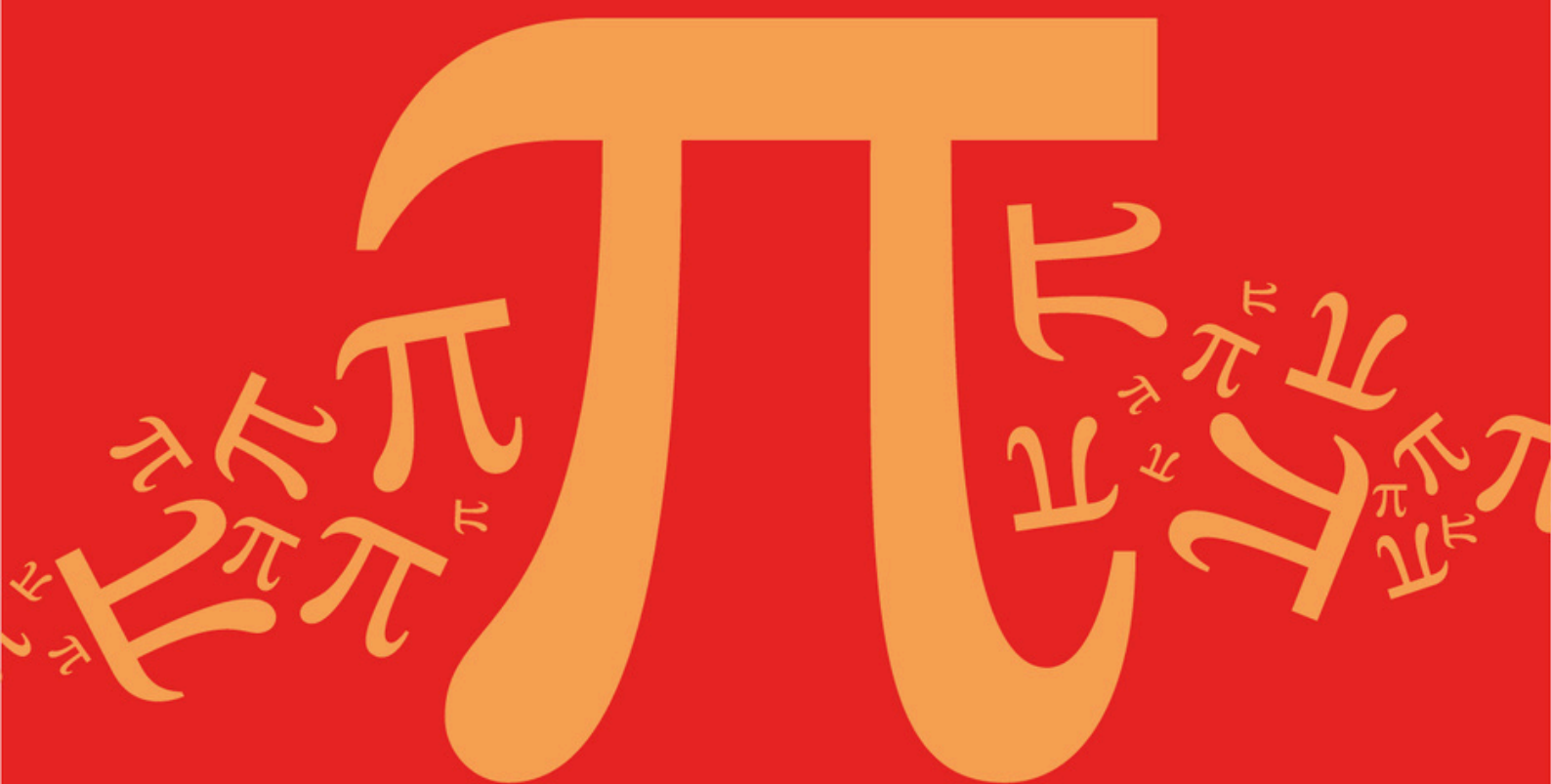
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DB: Networks

Sub topic

DB2: Network Optimisation/Spanning Trees

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

DB: Networks

Sub topic

DB1: Terminology for Networks

EXAM PAPERS PRACTICE

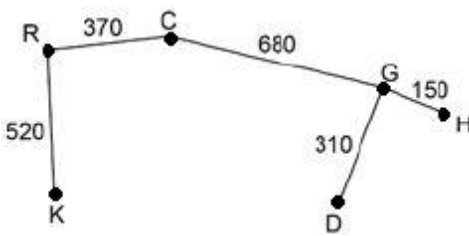
© 2024 Exams Papers Practice. All Rights Reserved

**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Sets up model and identifies one correct weight	AO3.3	M1	
Uses model to identify at least 4 correct weights Must identify no more than 5 weights	AO3.4	M1	
Identifies the correct minimum spanning tree for the network, stating clearly that it is a minimum spanning tree. Could be implied by clear evidence that Kruskal's algorithm or Prim's algorithm has been used.	AO1.1b	A1	Total distance = 2030 metres 2030×0.60 = £1218
Finds the correct total weight of the minimum spanning tree CAO PI	AO1.1b	A1	
Uses their total weight to find their correct minimum cost with consistent unit	AO2.2a	A1F	
			Total 5 marks

©2025 Exam Papers Practice. All rights reserved.

Q2.

(a) (i)

<i>AB</i>	6.1
<i>BC</i>	7.4
<i>BE</i>	9.7
<i>DE</i>	7.2
<i>EF</i>	10.6
<i>EH</i>	12.5
<i>HI</i>	6.7
<i>GH</i>	8.9

Prim's, 1st 3 correct, must be edges not lengths and no cycles

M1

8 edges

B1

EF 5th

A1

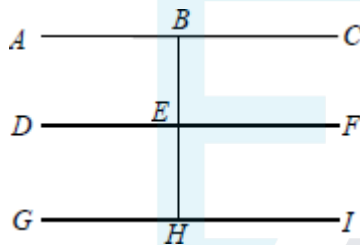
All correct

A1

(ii) (Length $\Rightarrow$) 69.1

B1

(iii)



Spanning tree with 9 vertices and 8 edges

M1

All correct, including labelling

A1

7

(b) (i) GH

B1

(ii) EF

B1

2

(c) (i) 1st AB

B1

(ii) Last EH

B1

2

[11]

Q3.

(a) (i)

EG	$\left(\begin{array}{c} 2.3 \\ 2.5 \\ 2.9 \\ 3.1 \\ 3.2 \\ 3.4 \\ 3.6 \\ 3.9 \\ 5.4 \end{array} \right)$
AB	
IJ	
AC	
AD	
HJ	
GJ	
BE	
FI	

SCA, Kruskal's, 1st 3 edges correct, must be edges not lengths, and no cycle in solution

9 edges

AD 5th

All correct

M1

B1

A1

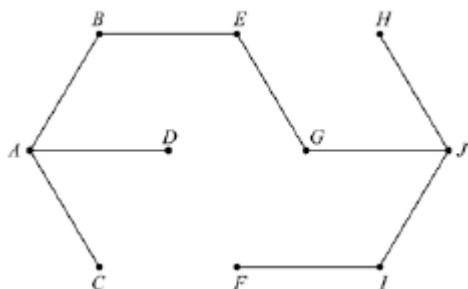
A1

(ii) 30.3

B1

©2025 Exam Papers Practice. All rights reserved.

(iii)



Spanning tree with 10 vertices and 9 edges.

M1

All correct including labelling

A1

7

(b) (i) FI

B1

(ii) DA

B1

2

[9]

Q4.

(a)

$$\begin{array}{l}
 ED = 6 \\
 AC = 8 \\
 AD = 10 \\
 \text{or} \\
 DC = 10 \\
 FG = 11 \\
 BE = 12 \\
 CF = 16
 \end{array}
 \left. \vphantom{\begin{array}{l} ED = 6 \\ AC = 8 \\ AD = 10 \\ \text{or} \\ DC = 10 \\ FG = 11 \\ BE = 12 \\ CF = 16 \end{array}} \right\}$$

Kruskal, must have first 2 edges correct & no cycles
(edges **not** lengths must be seen)

M1

AD or CD third edge

A1

BE 5th edge

A1

6 edges

B1

All correct

©2025 Exam Papers Practice. All rights reserved.

A1

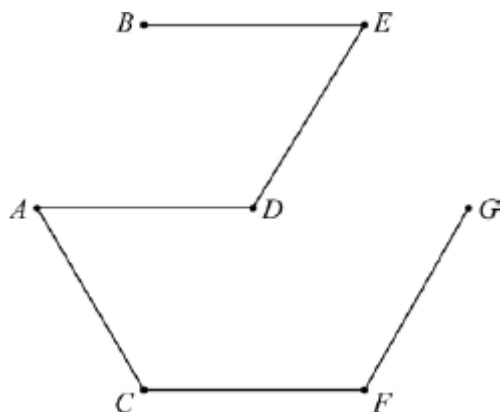
5

(b) 63

B1

1

(c)

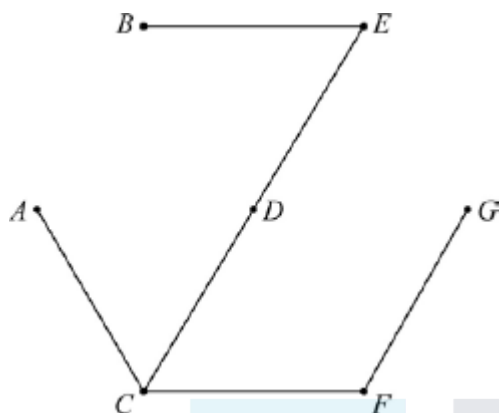


Spanning tree with 5+ edges

M1

Correct including labelling

A1



Correct including labelling on a separate diagram

A1

3

[9]

Q5.

(a)

	A	B	C	D	E	F	G
A					8		
B					7		
C					13		
D					4		
E	8	7	13	4	—	10	19
F					10		
G					19		

5 correct values in an E 'line'

B1

All correct

B1

2

(b) (i) **BADEFGCB**

Tour visiting at least 6 vertices

M1

Visits all 7 vertices

A1

Correct order from B

A1

B1

4

(ii) $B A D E F G \underline{E} C \underline{A} B$

Expansion of GC or CB

M1

Both correct

A1

2

(iii) 76

Minimum of 76 and their (b)(i)

B1F

1

(c) (i)

	①	②	④	③	⑤	⑥
	A	B	C	D	F	G
A	-	2	6	4	16	27
B	②	-	8	3	15	26
C	⑥	8	-	10	23	32
D	4	③	10	-	12	23
F	16	15	23	⑫	-	20
G	27	26	32	23	⑳	

Use of matrix form, 4+ numbers circled and 4+ parallel 'lines' crossed out

M1

C added 4th

A1

Any 5 values 'circled'

B1

All correct values circled and lines crossed out, either as shown or as mirror image.

Order of vertices must be clearly shown.

Condoned omission of line at G.

A1

4

(ii) 43

For 43 seen, or for $2 + 6 + 3 + 12 + 20$

B1

$$43 + (4 + 7)$$

Their 43 + 2 different edges from E

M1

$$= 54$$

SC 54 with no working 2 / 3

A1

3

(iii) 64

B1

1

(d) $64 \leq T \leq 76$

Must be written in symbols

B1B1

2

[19]

Q6.

(a) (i)

<i>AD</i>	4
<i>AB</i>	6
<i>AC</i>	16
<i>DE</i>	19
<i>EG</i>	10
<i>GI</i>	12
<i>IH</i>	13
<i>IF</i>	17

Using Prim's, first 3 edges correct, 6+ edges, no cycles, must have edges not lengths

M1

8 edges

B1

GI 6th

A1

All correct

A1

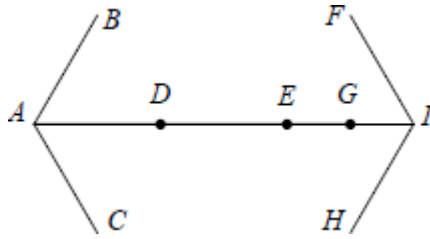
4

(ii) 97

B1

1

(iii)



ST with 6+ edges

All correct including labels

M1

A1

2

(b) (i) *IF*

B1

1

(ii) *AC*

B1

1

[9]

Q7.

(a) (i)

<i>EB</i>	(5)
<i>EH</i>	7
<i>AB</i>	8
<i>HI</i>	9
<i>AD</i>	10
<i>DG</i>	4
<i>EF</i>	12
<i>FC</i>	6

Prim's, MST, 6+ edges (no cycles), edges not lengths or vertices, with first 2 edges correct

M1

8 edges

B1

AB 3rd

A1

All correct

A1

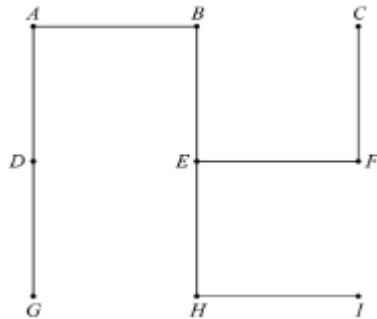
4

(ii) 61

B1

1

(iii)



6+ edges, connected, no cycles

Correct, including labelling

M1

A1

2

- (b) Delete BA , BE and reconnect with 1 edge
or
a spanning tree with 7 edges not including B
(either as a list or diagram)

P1 from their diagram in (iii)

M1

$$(61 - 13 + 11) = 59$$

Note: 59 scores 2/2

A1

2

[9]

Q8.

- (a) (i) AC
 CH
 FH
 CE
 CD (or ED)
 GH
 DB

Prim's, ST, 5+ edges (no cycles), edges not lengths or vertices, with first 4 edges correct

M1

7 edges

B1

CD (or ED) 5th

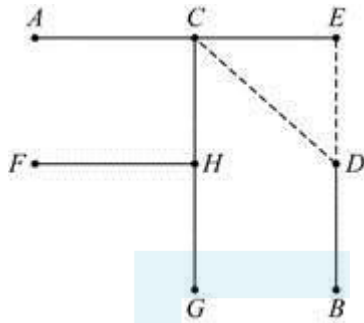
A1

All correct

A1

4

(ii)



CD, ED either of these lines
ST with 5+ edges, connected, no cycles

M1

Correct, including labelling

A1

2

(iii) 75(p)

B1

1

- (b) Delete CH, HG, HF and add FA and one of GC, GA, GD, GF

Deleting their edges connected to H , and
adding edges to make a ST with 6 edges

M1

or
a ST with 6 edges not including H
(either as a list or a diagram)

70(p)

Note: 70 scores 2/2

A1

2

[9]

Q9.

(a) (i)

AC	13
AE	14

<i>EI</i>	15
<i>CD</i>	16
<i>CH</i>	20
<i>EF</i>	21
<i>FB</i>	19
<i>BG</i>	19

Use of Prim's (not Kruskal's and not path); 6+ edges (no cycles); edges, not lengths or vertices, with first 2 edges correct

8 edges

CH 5th

EF 6th

All correct

M1

B1

A1

A1

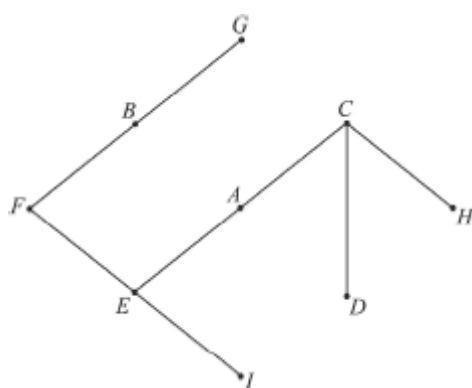
A1

5

(ii) 137

B1

1



6+ edges, no cycles

M1

Correct, including labelling

A1

2

(b) (Odds) *B, C, D, E*
PI CAO

E1

$$BC + DE = 22 + 18 \text{ (or 40)}$$

$$BD + CE = 38 + 27 \text{ (or 65)}$$

$$BE + CD = 22 + 16 \text{ (or 38)}$$

3 correct sets of pairs (lettered)

M1

3 correct sets of numbers; 2 correct sets of numbers

A2; 1

$$\text{min} = 307 + 38$$

PI 307 plus their shortest

$$= 345$$

A1F

B1

6

SC:

345 with no M mark scored scores 2/last 5

Route without 345 scores 0/last 5

[14]

Q10.

(a)

HI	6
DE	8
IJ	9
IG	11
AB	12
CG	14
BF	16
BE	17
FI	19

Kruskal's, 6 + edges stated, not just lengths, (no cycles) must be in ascending order (condone 1 slip only)

M1

9 edges

B1

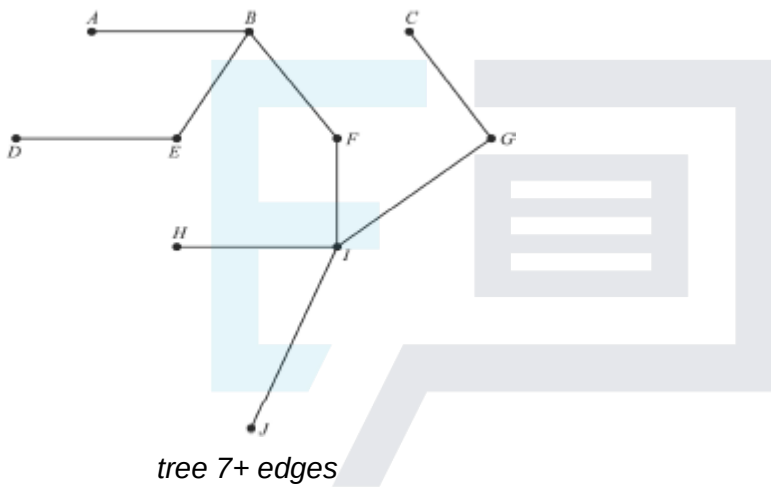
IJ 3rd

A1

AB 5th

	<i>BF 7th</i>	A1	
	<i>All correct</i>	A1	
		A1	6
(b)	112		
		B1	1

(c)



9 edges M1
 ©2025 Exam Papers Practice. All rights reserved. A1
All correct, including labelling

		A1	3
(d)	CG		
		B1	1

[11]

Q11.

(a)

- GH (5)
- GE (7)
- HJ (8)
- BE (10)
- BD (11)
- IH (14)
- DC (15)

AC (6)
 FJ (19)
 HK (22)

SCA allow Prim's from any vertex
but not Kruskal or path – min of 8 edges

10 edges	M1
HJ 3rd	B1
BE 4th	A1
AC 8th	A1
All correct	A1

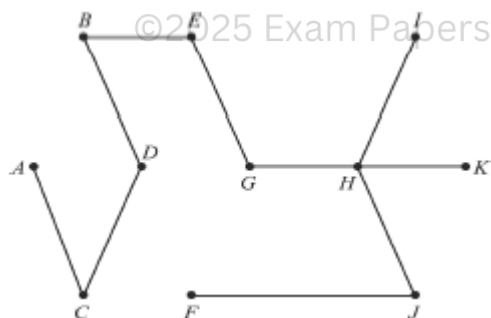
6

(b) 117

B1

1

(c)



(Possibly shown in part (a))
 MST (8+ edges)

10 edges	M1
All correct (+ vertices labelled)	A1

A1

3

[10]

Q12.

(a) Min MST

4 edges

M1

$$= 8 + 10 + 10 + 11 = 39$$

A1

2

(b) Max MST = $8 + 17 + 17 + 18$

$8 + 18 + 2 \text{ others}$

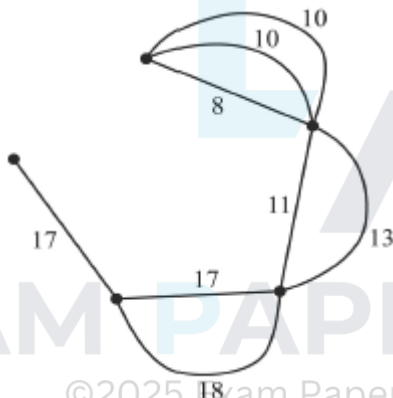
M1

$$= 60$$

A1

2

(c)



Connected graph with 5 vertices (all edges numbered, from G)

M1

$$MST = 53$$

$8, 11, 17, 17$ or $8, 10, 17, 18$

A1

other edges OE

A1

(other possibilities not shown)

(all edges numbered, from G)

3

[7]

Q13.

(a) (i) 9

B1 1

(ii) $n - 1$

B1 1

(b) (i)

EF	8
BC	8.5
CG	10
JI	11.5
BI	12
AB	14
GE	16
CH	16.5
DE	21

SCA minimum spanning tree, 7+ **edges**
(not cycles), must be in ascending order
and edges required (not lengths alone)

M1

BC 2nd

A1

JI 4th

A1

©2025 Exam Papers Practice. All rights reserved.
9 **edges** (not lengths alone) – may be
earned in (b)(iii)

B1

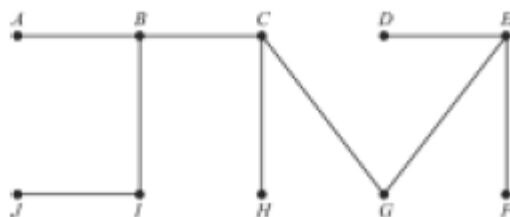
All correct

A1 5

(ii) 117.5

B1 1

(iii)



7+ edges, minimum spanning tree

Correct, including labelling

M1

A1

2

[10]

Q14.

(a)

DF	1.2
IH	1.8
BC	2.1
AJ	2.2
EF	2.4
HG	2.6
GF	2.7
AB	2.8
JI	2.9

or

9 edges

SCA

B1

AJ 4th

M1

©2025 Exam Papers Practice. All rights reserved.

A1

HG 6th

A1

All correct

A1

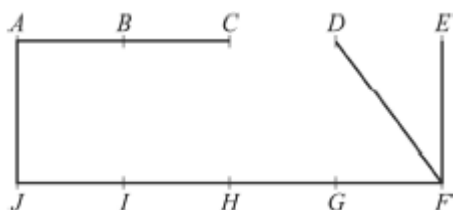
5

(b) 20.7

B1

1

(c)



MST – connected (7+ edges)

M1A1

2

- (d) EF (or 2.4)
for BC, DF, EF

M1A1

2

[10]

Q15.

- (a) (i) 10

B1

1

- (ii) $n - 1$

B1

1

- (b) Condone candidates attempting all of part (b) together / in different order

- (i) AB
 BC
 BD
 CF
 DG or FJ
 GK JK
 KJ GK
 KH or KI
 KI IE
 EI KH

Using Prim's

M1

BD 3rd

A1

CF 4th

A1

All correct

A1

10 edges

B1

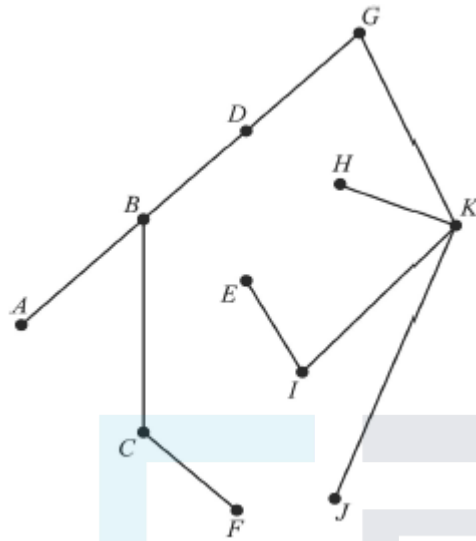
5

- (ii) (Length $\Rightarrow$) 155

B1

1

(iii)



Spanning tree with at least 8 edges
Any cycle scores M0

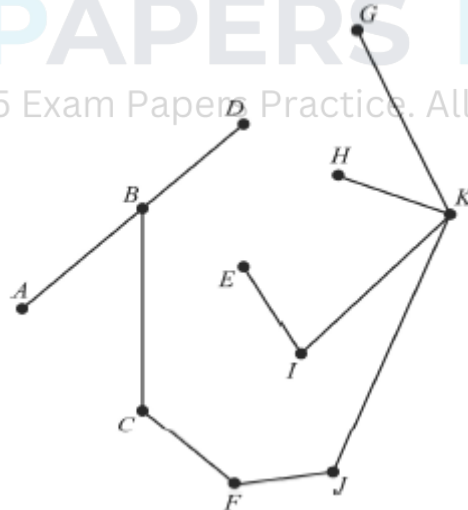
Correct and labelled

M1

A1

2

Alternative: FJ instead of DG:



©2025 Exam Papers Practice. All rights reserved.

[10]

Q16.

(a)

AB	5.5
BC	8
AI	9

BD	13
DE	9
DG	11
DF, EF, GF	12
IH	16.5

8 edges

B1

SCA

M1

AI 3rd

A1

BD 4th

A1

All correct

A1

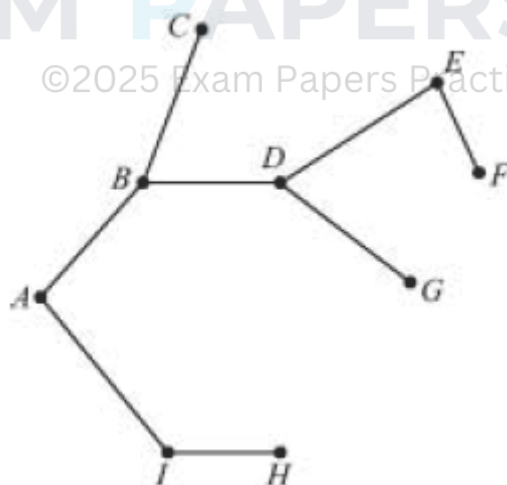
5

(b) 84

B1

1

(c)



Minimum spanning tree

M1

8 edges

B1

All correct including labelling (or including DF or GF instead of EF)

A1

3

(d) 2

B1

1

[10]

Q17.

(a) (i)

<i>SD</i>	12
<i>SC</i>	13
<i>SA</i>	14
<i>SB</i>	16
<i>DH</i>	75
<i>HG</i>	23
<i>GF</i>	22
<i>FE</i>	24
<i>EI</i>	81
<i>IJ</i>	12
<i>GK</i>	83
<i>KL</i>	16

Prim's (first 4 edges, allow 1 slip)

12 edges

HG 6th

EI 9th

All correct

M1

B1

A1

A1

B1

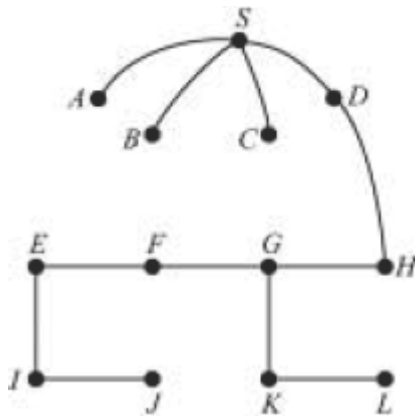
5

(ii) 391

B1

1

(iii)



MST (10 + edges)

12 edges

All correct

(iv) GF 7th (22)

HG 8th (23)

M1

A1

A1

3

B1

B1

2

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

(b) Odd vertices (E, H, J, K)

PI

E1

$$EH + JK = 69 + 131 = (200)$$

2 correct sets of pairings

M1

$$EJ + HK = 93 + 106 = (199)$$

$$EK + JH = 129 + 142 = (271)$$

A3,2,1,0

Repeat $EJ + HK$

$$\text{Total } 1135 + 199 = 1334$$

B1

6

[17]

Q18.

(a) (i) 9

B1
1

(ii) $n - 1$

B1
1

(b) (i) GI 5
9 edges

B1

AB 6
SCA
 EI 7
start with GI

M1

BD 8

A1

~~EG~~ 9

IJ 10
 IJ fifth

©2025 Exam Papers Practice. All rights reserved.

A1

HJ 11

~~HI~~ 12

111 12,

AF 13

DE 14

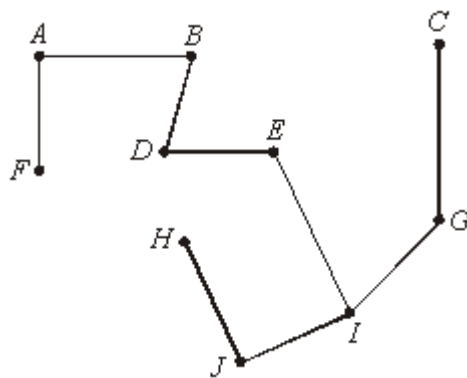
CG 15
all correct

A1
5

(ii) 89

B1
1

(iii)



9 edges

M1

A1

2

[10]

Q19.

(a)

AB	5
BD	3
DC	1
DE	4
DF	5
FG	6
GI	10
GH	11
HJ	13

SCA

9 edges

M1

B1

DC 3rd

A1

DE 4th

A1

All correct

B1

5

(b) 58

B1

1

[6]

Q20.

(a)	AB	3
	BC	6
	BE	13
	EF	5
	FD	or 10
	FG	32
	GJ	7
	GH	8
	HK	4
	HI	12

SCA Kruskal's (no method)

M1

(a)

B1

BE third

(b)

©2025 Exam Papers Practice. All rights reserved.

B1

A1

(c)

M1 A2

10 edges

B1

All correct

A1

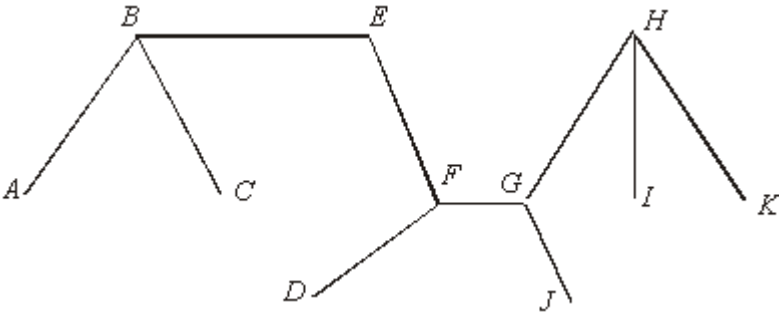
4

(b) $\mathcal{E} = 100$

B1

1

(c)



10 edges

(-1 EE)

M1

A2

3

(d) Seventh DF

B1

Eighth HI

B1

2

[10]

Q21.

- (a) AB or 20
AC 25
BD 30
CE 35
EF 40
FI 35
HI 30
IK 35
HG 40
HJ 45

SCA

M1

10 edges

B1

BD third

A1

CE fourth

A1

All correct

A1

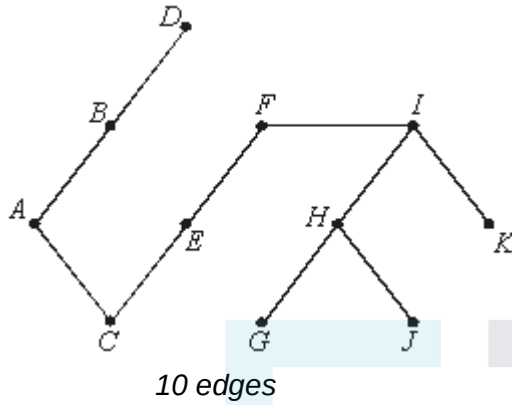
5

(b) 335

B1

1

(c)



10 edges

B1 M1 A1

3

(d) Add AE

+40

Adding AE, deleting CE/CA

M1

Delete CE
Extra

-35

+5

15 with no working (M1, A0)

A1

2

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[11]

Q22.

(a)	AB	8
	IH	9
	CD	10
	EF	11
	BE	13
	BC	14
	(not AF)	
	(not DE)	
	DI	20
	HG	22
	SCA	

M1

HI second

A1

SC (Prims Max 3/6)

1 (a) B1 8 edges

1 (b) B1 107

1 (c) B1 MST
May be implied

All correct

A1

A1

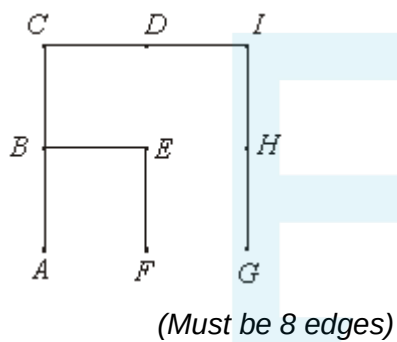
4

(b) Total 107

B1

1

(c)



B1F

1

[6]

Q23.

(a) $AB = 3$

SCA

M1

$FG = 4$

FG second

A1

$ED = 5$

$FC = 6$

$BD = 7$ or $AD = 7$

$CD = 9$

6 edges

B1

all correct

A1

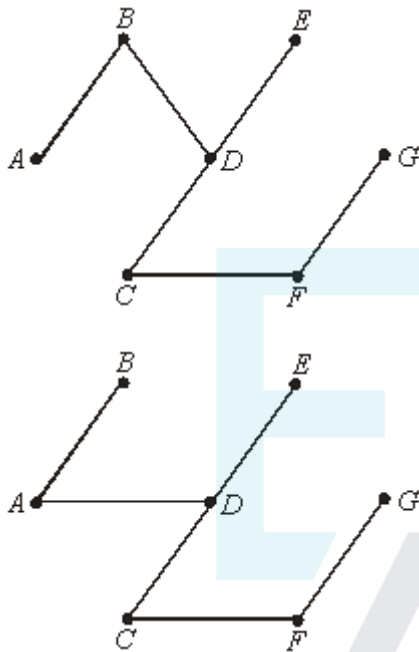
4

(b) Length = 34

B1

1

(c)



B1B1F

2

EXAM PAPERS PRACTICE [7]

©2025 Exam Papers Practice. All rights reserved.

Q24.

(a)	(i)	BC	—	6
		CD	—	8
		CE	—	9
		FH	—	9
		GH	—	10
		AC	—	11
		GI	—	12
		DF	—	14
		AM	—	30
				109

B1

SCA

M1*

9 edges
 *independent
 CE FH either order

M1*

Not DE

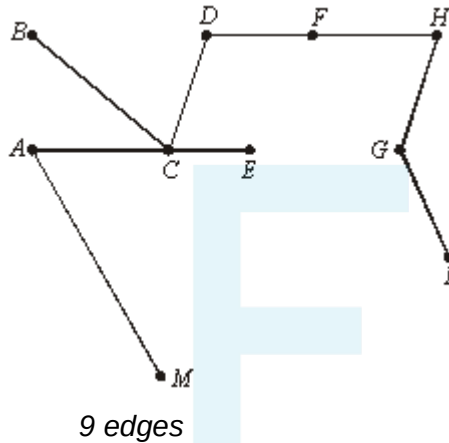
A1*

All correct
** Must have M2*

A1*

5

(ii)



M1

A1

2

(b) Odd vertices
 AB, HI

Min connect
 AB = 17
 HI = 22

B1

Reject AH, BI and AI, BH
or their values shown
(their total + their 39)

E1

Total 264 + 39
 = 303 metres

M1

CAO
 SC: 248 with no working 2/4
 SC: MR with ABHIM: M1 repeat HB
 A1 309

A1

4

[11]

Q25.

(a)

$R \rightarrow Y$	22
$Y \rightarrow L$	24
$R \rightarrow M$	31
$M \rightarrow S$	15
$S \rightarrow D$	11
$S \rightarrow N$	12
$D \rightarrow P$	56

Total 171

SCA

7 edges

YL second

All correct order

B1

5

M1

B1

A1

Kruskals/None

M0

B1

A0

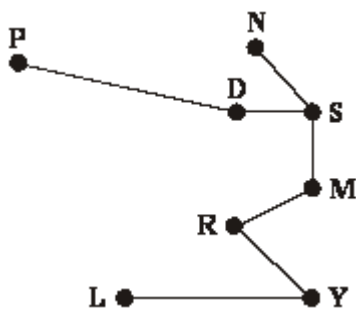
B1

B1F

3/6

A1

(b)



B1F

1

[6]

Q26.

(a) (i) $(ABDK) = 3$

B1
1

(ii) $MST = (n - 1)$

M1

$= 15$

A1
2

(b) (i) Odd vertices A and P

Route length = all edges + repeated edges (answer > 30)

M1 M1

Route length = all + (3)

A1

$(33 + 3) = 36$

B1
4

(ii) Repeated: AE, EL, LP

B1
1

©2025 Exam Papers Practice. All rights reserved.

(iii) E and L now odd

M1

Their $(33) - 2 + 1 = 32$

A1F
2

[10]

Q27.

FG 1

SCA

M1

BD 2

Start FG

A1

$\begin{matrix} DC & 3 \\ EG & 3 \end{matrix}$

Either order

A1F

$HJ \quad 4$

$\begin{matrix} AD & 5 \\ JI & 5 \end{matrix}$

Either order

A1F

$HG \quad 6$

$CF \quad 7$

Length 36

*SC: No method – max 3/6
 (Graph of 9 edges and of length 36)*

B1



©2025 Exam Papers Practice. All rights reserved.

B1

[6]

Q28.

(a) 1) AC 6

SCA (for MST)

M1

2) AE 7

Second

A1

3) EG 7

8 edges

M1

4) GD 1

5) DB 4

DB fifth

A1

6) GI 6

7) IH 7

8) FC 8

All correct

A1

Length = 46

[S.C. Using Kruskal's/no working, maximum 3/6.]

B1

6

- (b) (i) Odd vertices A & I
May be implied

M1

Min connector is

ABDGI = 19

Route / Distance

A1

Total length = 19 + 121

Total + their repeat

©2020 Exam Papers Practice. All rights reserved.

M1

(= 140)

Tour, 21 vertices

(3A, 3B, 2C, 2D, 3E, 1F, 3G, 2H, 2I)

B1

4

- (ii) Length = 140

B1

1

[11]

Q29.

- 1 BD
- 2 FE
- 3 HG
- 4 BC
- 5 AB

- 6 FG
7 DE

s.c.a

M1

start with BD

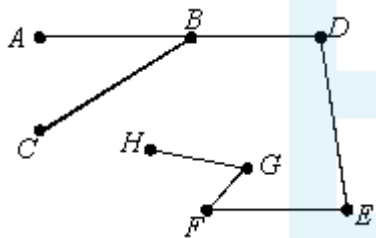
A1

FE second

A1

HG third

A1



correct tree

s.c: Prim's or no method scores up to 4

B1

Length = 5.15

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

B1

[6]

Q30.

(a) 3

B1

1

(b) (i) $n - 1$

B1

1

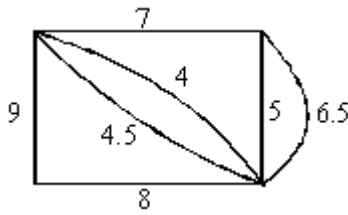
(ii) $(n - 1)!$

M1 for factorial

M1A1

2

(c)



connected graph

M1

m.s.t. = 17

M1

all other arcs used and correct

A1

A1

3

[7]

Q31.

- (a) Each row $7 \times 49 = 343$
 $1 \times 14 = 14$
 Row $7 \times 2 = 14$
 Mairs 5
 376

(O.E.)

M1

M1

All correct answers given

A1

3

- (b) $29 + 5 = 34$

B1

1

- (c) Rows $m(n - 1)$
Either columns or rows

M1

Column $(m - 1) \times 2$

A1

Total $m(n - 1) + 2(m - 1) + 5$
 $= mn + m + 3$

A1cao

3

(d) Rows $\frac{x}{60}$

Or using answer to part (c)

Total $59 \times \frac{x}{60} + \left(\frac{x}{60} - 1 \right) \times 2 + 5$

M1

$= \frac{61x}{60} + 3$

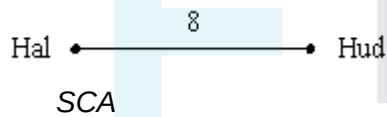
A1cao

2

[9]

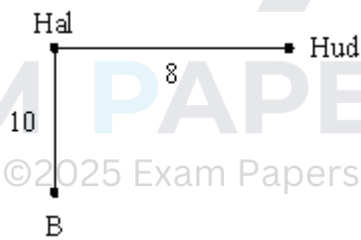
Q32.

(a) (i)



M1

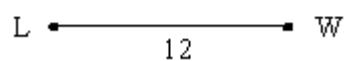
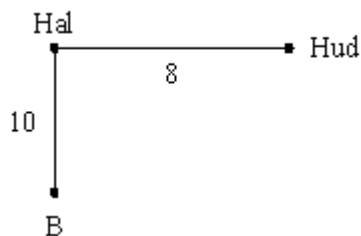
(ii)



Must be Hal/Hud

A1

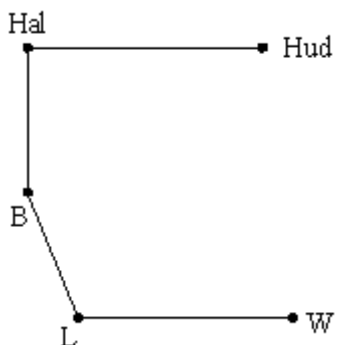
(iii)



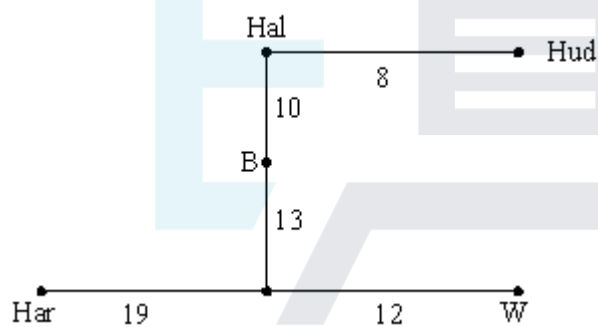
Not Hud

A1

(iv)



(v)



A1, CAO

4

(b) 62

©2025 Exam Papers Practice. All rights reserved.

B1

1

[5]

Q33.

(a) (i) Need $4 + 3 + 2 + 1$ edges

E1

1

(b) (i) M.S.T. = $7 + 7 + 7 + 9 = 30$

B1

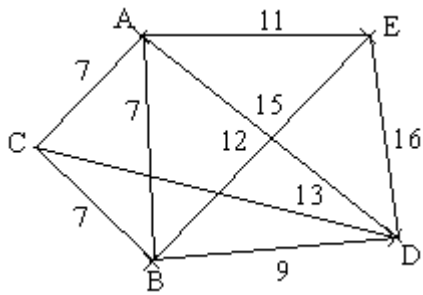
1

(ii) Might include loop

E1

1

(iii)



Must include loop with

7's

M1

M1

No loop with 9 & 11

A1

3

[6]

Q34.

(a) AB (3)

Overall method

M1

5

BC (4)

A1

©2025 Exam Papers Practice. All rights reserved.

BD or CD (7)

Omitting loop

A1

CE (10)

CE

A1

EF (9)

All correct

A1cao

(b) 33

B1

1

(c) Easier to detect loops

E1

1

[7]

Q35.

- (a) BD, BF
(7) (8)

M1

Evidence of correct method

A1

BA CF BE
(12) (12) (13)

M1

Avoiding DF

A1

4

- (b) 52

[s.c. using Prims or inspection 3/5]

B1F

1

[5]

Q36.

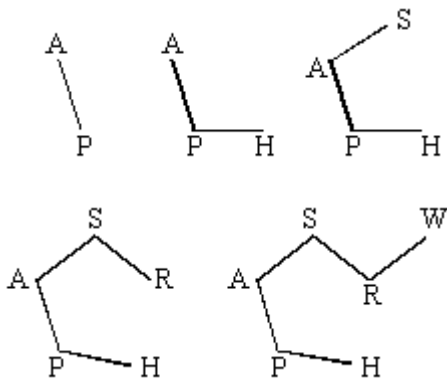
- (a) $(n - 1)$

©2025 Exam Papers Practice. All rights reserved.

B1

1

- (b)



Prior's???

M1

2nd sketch

A1

3rd sketch

A1F

4th sketch

A1F

All correct

A1cao

5

(c) MST = 15

B1

1

[7]

Q37.

(a) PQ, PT 30, 30

TU 30

ST 35

M1

A1

A1

QR 40
UV 50

©2025 Exam Papers Practice. All rights reserved.

A1

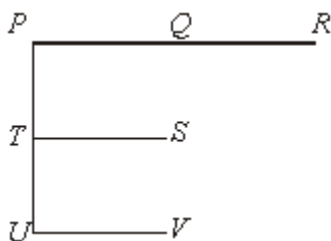
Total cost 215p

For cao: B2

B1

5

(b)



M1 A1ft

2

(c) PQ allowed but QR not

M1

Cheapest alternative to reach R is by SR
raising cost by 5p to 220p

A1ft

2

[9]

Q38.

(a) AF 10

EF 10

DE 15

CF 25

AB 20

M1 A1

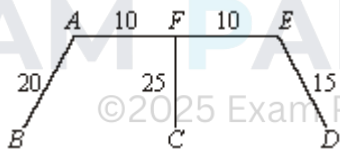
A1

A1

A1

5

(b)



(numbers not needed until part (c))

M1 A1

2

(c) All points are linked by edges in the spanning tree with maximal paths

M1

$20 + 10 + 25$, $20 + 10 + 10 + 15$ and

A1

$25 + 10 + 15$ all less than 60 minutes

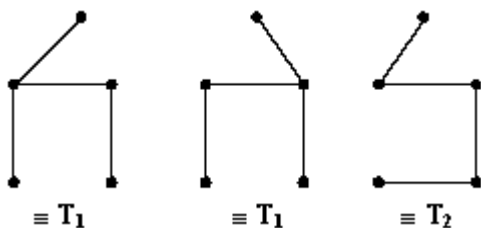
A1

3

[10]

Q39.

(a)



(for subgraphs)

M1

A1× 3

6

(b)

fit sensible answers

B1× 3

3

[9]

Q40.

(a) AC 17 AE 18

AD 19 BC 20

BF 21

M1/A1

A1

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Total length 95 miles

B1

5

(b) AC, AE, AD as before

M1

Then AF (22) and BF, increasing total by

2

A1

A1

3

[8]

Q41.

(a) e.g. AB 5, AG 7, GF 6, FE 5, ED 5, GC 6

*Prim implied by work on table
or by order of choice*

M1 A1 A1 A1

total length 34

B1

5

- (b) maximum tolled = complement of
minimum connector

M1 A1

2

*these two marks can be earned by any
sensible alternative approach*

length = $117 - 34 = 83$

A1

- (c) Now the connector must include BG (15).

M1

The most economical way of adapting
the minimum connector is to replace

AG (7) by BG .

*these three marks can be earned by any sensible
alternative approach*

A1

So new minimum is 8 more and new max

tolled = $83 - 8 = 75$.

('15' anywhere earns the M1)

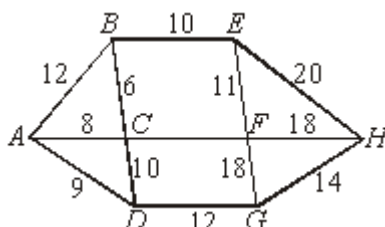
B1

3

[10]

Q42.

(a)



M1 A1 2

(b)

M1

BC, AC, AD,
lowest edges

A1

BE, CF (or vice-versa),
Selected 10s

A1

DG and GH
Rest

A1

Total cost = $6 + 8 + 9 + 10 + 10 + 12 + 14 = £69$

B1

5

(c) (i) The four numbers in C's row should be halved

B1

(ii) A sends information: lowest *AC* as *before*

©2025 Exam Papers Practice. All rights reserved.

M1ft

Use *CB, CF* as before

A1

Use *CD* instead of *AD*

A1

New lowest cost = $3 + 8 + 5 + 10 + 5 + 12 + 14 = £57$

B1

5

[12]

Q43.

(a) *AD* 5

Start

M1

2

AF 6

A1

FG 6

A1

AE 7

Dealing with the 7s

M1

2

EC, EH, BC (all 7) in some appropriate order

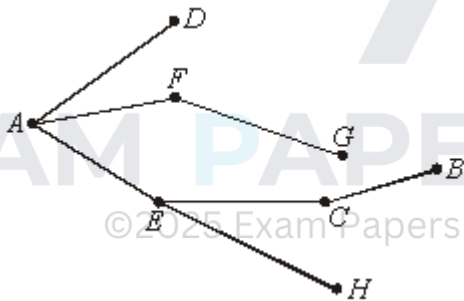
Total 45

Or matrix method

M1

2

(b)



M1 A1

$$AECB = 7 + 7 + 21$$

B1ft

3

(c) A to B too long

B1

Need to grit BD instead of BC

M1

(adding just one extra mile to the gritting)

A1

3

[12]

Examiner reports

Q1.

Over 75% of students were able to score 3 or more marks for this question, which is very positive for this type of unscaffolded, new-style question. Marks were lost through numerical errors, selecting incorrect edges in the minimum spanning tree and not being explicit about which algorithm was being used to solve the problem.

Q2.

The three sections of part (a) were invariably correctly answered and clearly presented. Part (b) proved to be more challenging. Students were asked for edges but 'G' and then 'F' were quite common as answers. Part (c) was well answered with very few incorrect answers.

Q3.

Generally this was a very high scoring question with many students scoring full marks. The three parts of part (a) were usually shown clearly and calculated accurately. Many students are clearly taught to write down all the sides in order of ascending length and then to tick or cross them according as to whether they are to be included, without writing the resulting spanning tree out separately. Very few students forgot to label their diagram. In the second part, a few students confused the request for edges with one for vertices.

Q4.

The question was almost always very well answered. Only a small minority failed to make the order of selection of edges in part (a) clear or label their diagrams in part (b).

Q5.

©2025 Exam Papers Practice. All rights reserved.

There were very few errors in part (a) which was set in order to give candidates the opportunity to 'see' the context of the question.

In part (b)(i) the candidates who elected to write their tour down, vertex by vertex, underneath the table usually earned full marks. Those who elected to work entirely in the table often failed to make clear the order in which vertices had been selected and thus that they had actually used the nearest neighbour algorithm.

Despite answering part (b)(i) correctly and the hint in part (a), a surprising number of candidates merely repeated their tour from part (b)(i) for part (b)(ii).

Part (b)(iii) was usually correct.

Part (c)(i) was poorly answered. Many candidates simply did not present their working "on Table 2" and of those that did, few showed the order of selection of edges.

Part (c)(ii) caused even more problems. Many thought the required lower bound for the length of the tour would be the same as the length of the spanning tree just found. A larger number appeared to misread the requirement completely and now deleted a vertex from Table 2, going on to find a lower bound for a tour of the vertices in Table 2.

The last two parts of the question, part (c)(iii) and part (d), were quite well answered by the majority of candidates.

Q6.

The question was almost always very well answered. In part (b) weaker students often confused 'edge' and 'vertex'.

Q7.

Generally this was a very high scoring question with many full marks gained. Even candidates who earned very few marks elsewhere on the paper usually managed a high mark.

In part (a) it was pleasing that the answer to part (a)(i) was more clearly presented, from almost all candidates, than a similar answer on last summer's paper. Very few forgot to label their spanning tree in part (a)(iii).

Part (b) presented no problem at all. Most subtracted two lengths and added an acceptable replacement - really the only error in this part was to add the wrong length.

A considerable minority reworked the question from scratch with B removed.

Q8.

This question was a problem for some candidates, with their failure to understand Prim's or Kruskal's algorithms and their persistence in finding a route. However, for the better candidates, the question was generally well answered. It was apparent that candidates had attempted a correct application of Prim's algorithm but totally failed to indicate the **edges** involved correctly. These candidates had often done all their working within the table of the question. Candidates should be aware that even where they have attempted to show the order of choice of vertices within the table, it is not always clear to examiners precisely which edges are involved. Whatever method of working, there should be an explicit statement of the order of choice of vertices.

In part (b), many candidates deleted only two edges from their minimum spanning tree instead of the required three. Others failed totally to make clear which edges were being deleted and added, offering only a sequence of apparently random numbers which was futile given the number of repeat values in the table.

Q9.

Most candidates scored full marks or, at worst, dropped 1 mark: $BD + CE$ in part (b).

Q10.

Generally this was a very high scoring question with many candidates scoring full marks. The most common error was to choose the wrong edge in 8th or 9th place in part (a).

In part (a) the preferred method of many candidates was to write down all edges in order of increasing size and then tick or cross them to indicate selection or rejection. The candidate's answer was almost always clear but it would have been nice if all candidates were encouraged to write out their final answer explicitly.

In part (d) a very common error was to confuse the terms 'edge' and 'point/town' and offer C as the answer.

Q11.

Part (a) was reasonably well answered, although a significant number of candidates made

minor slips, a reflection perhaps of the complexity of the diagram. Presentation continues to improve but a few candidates lost marks because it was not possible to decipher their method.

In part (b), very few candidates failed to label their diagram.

Q12.

The usual mark scored was 4 out of 7, with very few candidates getting part (b) or the final mark in part (c) due to adding in extra edges and changing the MST.

Part (a) was usually correct.

Part (b) was usually incorrect, with $13 + 17 + 17 + 18 = 65$ being the almost universal answer. The very few who answered correctly were by no means always drawn from those scoring highest on the rest of the paper.

In part (c), most of the candidature spotted one of the two correct MSTs. A number of candidates elected to draw only a correct MST. Those candidates who attempted to complete the diagram frequently forfeited marks as their extra edges impacted on their MST. A few candidates drew separate diagrams.

Q13.

Part (a)(i) was almost always correct; likewise for part (a)(ii), although a number of candidates gave their answers as 11 and $n + 1$. Part (b) was usually correct, apart from the occasional slip with the 8th edge. Very few candidates failed to make clear their method at all. They were probably equal in number to those clearly using another method.

Q14.

This question was well answered by the majority of candidates, often yielding full marks, even to otherwise weak candidates. Presentation was improved compared to equivalent questions from previous series, and only the weakest candidates lost marks through poorly presented work in part (a).

Q15.

Both parts of part (a) were almost always correct.

In part (b), the great majority used Prim's algorithm although a small but significant minority still do not make it clear from their working that Prim's is actually being used. The order of selection of edges must be clear. A few candidates started from G and a considerable number worked successfully until the 6th or 7th edge. The edge BC was often missed for a long time in otherwise correct work. Almost all candidates succeeded in drawing their minimum spanning tree and correctly adding up its length.

Q16.

Candidates must list the order in which edges are added to their spanning tree. A significant number of candidates started using the correct algorithm but failed to persevere to the end of the question.

Q17.

There is still a minority of candidates who fail to make clear the order of edge selection,

but generally the first three parts were well done, although a few failed to label the minimum spanning tree. Part (a)(iv) was often incorrect, with finding the seventh and eighth edges proving to be difficult for a significant number of candidates. Edge BC appeared regularly.

Responses to part (b) showed a real advance over previous papers, with many fully correct and well-presented solutions. The Chinese postman was usually well done, with the vast majority identifying the three possibilities. A small number had 6 odd nodes even though the specification states that there will be no more than 4. It was encouraging to see that the given total was used and there was only a small number who wanted to add up all the edges.

Q18.

This question was a good source of marks for virtually all candidates, with few scripts using the wrong algorithm. Candidates who listed all the edges in part (b) were successful, although a common mistake was including edge FH . Some diagrams of minimum spanning trees were not labelled at the vertices.

Q19.

This question was a good source of marks for virtually all candidates, with few scripts using the wrong algorithm. A significant number of candidates merely drew the correct spanning tree, with no working. These candidates were rewarded, but not with full marks. It is essential that **all** working is shown if a candidate is to score full marks. Some diagrams of minimum spanning trees were not labelled at the vertices.

Q20.

This question was a good source of marks for the majority of candidates. There was a number of candidates who drew a spanning tree in part (c) without labelling their tree. In part (d), candidates were expected to use their previous working to find required edges. However, a large number of candidates started from scratch and applied Kruskal's algorithm to the network.

Q21.

Many candidates failed to indicate that Prim's algorithm had been used. Candidates must list the order in which edges are added to their spanning tree. Also a significant number of candidates started using the correct algorithm but failed to persevere to the end of the question with a loss of marks ensuing.

Q22.

This question was a good source of marks for the majority of candidates. A number of candidates drew a spanning tree in the final part without labelling it.

Q23.

Many candidates failed to indicate that Kruskal's algorithm had been used. Candidates must list the order in which edges are added to their spanning tree. There were a surprising number of candidates who scored full marks in parts (a) and (b) but failed to correctly identify the second minimum spanning tree.

Q24.

The majority of candidates performed well in part (a), although some weaker candidates thought that the minimum spanning tree should produce a full cycle.

Part (b) was poorly answered. Many candidates thought that this part was independent of part (a) and merely found the length of a Chinese Postman route for the four vertices mentioned in the table. Although this approach was fundamentally wrong, candidates still received some credit for their work on the odd vertices.

Q25.

Candidates who listed the arcs scored well on this question, although a common mistake was choosing DN rather than SN. Some candidates converted the diagram into a matrix before applying Prim's algorithm. Candidates need to be familiar with both versions of Prim's algorithm. If using the matrix method, candidates must indicate the order in which the arcs are added.

Q26.

Part (a) was well answered.

Part (b) proved to be more challenging. Candidates were able to make a good attempt at part (i), although a significant number used $16 + 2$ or 3 . In part (ii), a common mistake was to omit the edge EL . Part (iii) proved to be a good discriminator. Candidates failed to realise that by removing AE and LP , then E and L would become the odd vertices.

Q27.

This question was generally well done, except by the very weakest candidates, but mistakes were made on the drawing of the minimum spanning tree.

Q28.

Candidates scored high marks on part (a) of this question, although there was a significant number of scripts with careless slips. In part (b) candidates realised that the length of an optimal Chinese Postman route must include a repetition of A to I but did not use the minimum spanning tree found in part (a) to find the minimum additional distance.

Q29.

In general this question was answered well, although some candidates used Prim's algorithm. Those who failed to draw their final tree or state its length lost marks.

Q30.

Most candidates made a reasonable attempt at this question but very few scored full marks. Many did not see the connection between parts (a) and (b)(i). In part (c) many candidates were able to draw a correct minimum spanning tree but very few were able to add on the remaining arcs successfully.

Q31.

This question was a good discriminator with a number of candidates scoring full marks. Only the most able candidates were able to successfully manipulate the algebraic expression in part (d).

Q32.

This question was not answered as well as corresponding ones in previous years. Candidates did not show their working in part (a) and were penalised accordingly. They must know that Kruskal's algorithm adds the lowest edge to the tree, but this edge does not have to be connected to the tree. Some candidates had the minimum spanning tree with only four edges.

Q33.

This question proved to be discriminating between the candidates. Although all candidates were able to attempt all of the question there were very few correct answers to part (b)(iii).

Q34.

The first two parts of the question were well answered. However, weaker candidates did not show their working in part (a) and were penalised accordingly. Candidates should know the difference between Prim's and Kruskal's algorithm and their relative merits. However very few candidates were able to score the mark in part (c).

Q35.

The majority of candidates scored full marks on this question. Kruskal's algorithm requires choosing the shortest edge and adding the next shortest edge, which doesn't complete a circuit, until every vertex has been included. Candidates must show their working at each stage if they are to score full marks.

Q36.

This question was well answered with many scripts scoring full marks. Candidates who failed to score full marks often used Kruskal's algorithm. It is essential that all candidates are fully aware of the two algorithms, their differences and their relative merits.

©2025 Exam Papers Practice. All rights reserved.

Q38.

This was extremely well done.

Q39.

Candidates made very good attempts at constructing the three trees but, as one might expect, only the best were able to spot the isomorphic pairs.

Q41.

Most candidates found the minimum connector using the matrix method, and again the working shown on the insert was generally very clear. However, a few candidates gave the minimum connector as a list of vertices in the order in which they were included: that does not uniquely determine the connector, which should be given unambiguously as a set of edges.

In part (b) very few candidates realised that the maximum tolled roads are simply found by taking the complement of the minimum toll-free roads which essentially were found in part (a).

Q42.

All candidates could construct the network and they seemed to find the pre-printed vertices helpful. The vast majority could also construct a minimum connector, although a sizeable minority used Prim's algorithm rather than Kruskal's. The specification states that candidates should know both, and, when given a network, Kruskal's is marginally easier than Prim's. In adapting their answers for part (c) too many candidates simply adapted the costs in the same tree that they had constructed in (b) without realising that one edge needed changing too (CD now being cheaper than AD).



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.